

迈克尔逊干涉仪在压电陶瓷材料中的应用

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摘要: 本文分析了在迈克尔逊干涉仪基础上设计的测量压电陶瓷材料的逆压电系数以及由表面膜层形成迈克尔逊干涉条纹的原理及方法, 并介绍了它们的重要应用。

关键词: 迈克尔逊干涉仪; 压电陶瓷; 逆压电系数; 膜层

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1 前言

迈克尔逊干涉仪是根据光的干涉原理制成的精密测量仪器, 它可精密地测量长度及长度的微小变化率等。在现代科学技术中有着广泛的应用。压电陶瓷材料就是一种在电场作用下会产生伸缩效应, 且伸缩量极小的一种材料, 普通检测方式难以凑效。如将迈克尔逊干涉仪的反射镜 M 粘在压电陶瓷片上, 就可以测量出其逆压电系数。对于在基片上生长有透明介质膜层的厚度测定, 用一般方法测定也是困难的, 若将样品置于迈克尔逊干涉光路的一臂, 由膜层前后两表面间的反射光与来自干涉光路另一臂的反射光将产生两组干涉条纹, 根据干涉条纹即可测定膜层厚度, 具体介绍如下。

2 实验原理与方法

2.1 微振动

在图1所示迈克尔逊干涉仪中, P 为分束膜, M 为平面反射镜。将 M 粘在压电陶瓷片上, 当压电陶瓷片受到电激励产生机械振动时就带动 M 移动。由于在迈克尔逊干涉仪上, M 每移动 $\lambda/2$ 的距离就会产生一个光干涉的拍信号。因此根据光拍信号的个数就可以计算出压电陶瓷片伸缩的距离。现设压电陶瓷片上加有正弦激励电压为:

$$V(t) = V_m \sin 2\pi f_0 t$$

动镜 M 产生振动, 其位移为:

$$X(t) = KV_m \sin 2\pi f_0 t = A_m \sin 2\pi f_0 t \quad (1)$$

式中 K 为逆压电系数, A_m 为振动位移峰值。

对于光拍信号, 由于 M 在位移最小处速度最大, M 移动所花的时间最短, 光拍最密; 反之, M 在位移最大处

速度最小, 光拍最疏。可见, 随着时间的变化, 光拍信号是一个具有一定周期性的变频信号, 其周期为原振动信号周期的 $1/2$ 。对应于振动位移峰谷两处, 我们可在光拍信号的一个周期中读出光拍个数 N , 则 M 的振动位移峰值为:

$$A_m = N\lambda/4 \quad (2)$$

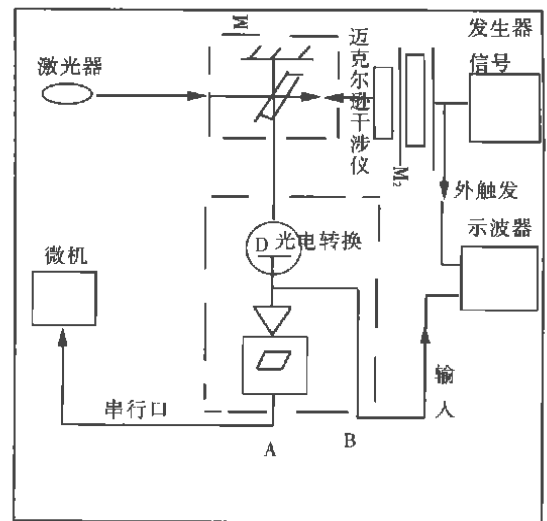


图1 (Fig. 1)

2.2 平行平面膜层

在图2所示平行光照射的迈克尔逊干涉光路中, M_1 为反射镜, M_1' 为 M_1 经 A 所成的像, p_1 和 p_2 为膜层的前后两表面, 膜层厚度均匀且为 d_0 , 折射率为 n , 设 M_1' 与 P_1 的夹角为 α , 交线坐标为 X_0 , 当 α 很小时, 在坐标为 X 处从膜层前表面的反射光与从 M_1' 的等效反射相干光的光程差为:

$$\Delta_1 = 2(X_0 - X)\alpha \quad (3)$$

THE WORLD OF REALITY KINDNESS AND BEAUTY

——The art character of Chinese folk china

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Abstract: Practicality with succinct and generous visual effect is the first character of folk china art. Another character of folk china is its unique beauty formed by its material and technic. Folk china art is the harmonization of combining the kind workers' pure hearts and their forms. The pure beauty of folk china art is the objectifying of the quality and style of manual labors.

Keywords: folk china; art character; practical; material and technic; pure beauty

(上接第 33 页 •Continued from page 33)

RELATIONSHIP ANALYSIS BETWEEN THERMAL EXPANSION AND X—RAY DIFFRACTION INTENSITY OF QUARTZ AND CRISTOBALITE IN ELECTRICAL PORCELAIN BODIES

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Abstract: Tests were carried out by measuring the thermal expansion and the X—ray diffraction intensities of high—tension insulators. The relationship between the X—ray intensity of quartz and cristobalite in electrical porcelain bodies and with their thermal expansion values at 650°C and 250°C was examined using scatter diagram and multiple linear regression analysis. A statistical relationship has been found between the thermal expansion and the relative X—ray peakheight intensities of quartz and cristobalite phases in electrical porcelain bodies.

Keywords: electrical porcelain; quartz; cristobalite; thermal expansion; X—ray diffraction intensity

(上接第 35 页 •Continued from page 35)

THE APPLICATION OF MICHAEL INTERNEE APPARATUS USED IN PIEZO CERAMICS MATERIAL

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Abstract: In this article, I analyze inverse piezoelectric coefficient which measures piezo ceramic material and which bases on the design of interference apparatus. I also analyze the principle and method of forming the Michael interference stripe on the surface of electronic film. Their important application is introduced too.